O-C-275C
June 30, 1969
SUPERSEDING
Int. Fed. Spec. O-C-00275B(GSA-FSS)
April 5, 1968 and
Fed. Spec. O-C-275
December 8, 1960

FEDERAL SPECIFICATION

CHEMICALS, PHOTOGRAPHIC "SINGLE SUBSTANCE", GENERAL SPECIFICATION FOR

This specification was approved by the Commissioner, Federal Supply Service General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers requirements and tests for chemicals used in photographic processing.

1.2 Classification.

1.2.1 Grade. This specification covers one grade of photographic chemicals designed as "photographic grade." (see 6.3)

1.2.2 Classes. The photographic chemicals covered by this specification shall be of the following classes:

Column 1	Column 2 USA Standards (See 2.2)
Class I - Acids	
Acetic Acid, Glacial	PH4.100
Sulfuric Acid	PH4.101
Citric Acid, Monohydrate	PH4.102
Boric Acid, Crystalline	PH4.103
Hydrochloric Acid	PH4.104
Sodium Acid Sulfate, Fused	PH4.105
Acetic Acid, 28-Percent Solution	PH4.106
Citric Acid, Annydrous	PH4.107
Class II - Developing agents	
p-Methylaminophenol Sulfate (Armol, Elon, Genol, Graphol, Metol, Photol, Pictol, Rhodol, Veritol, Verol)	PH4.125
Hydroquinone (Para-Dihydroxybenzene, Quinol, Hydroquinol)	PH4.126
2, 4-Diaminophenol Hydrochloride (Acrol, Amidol, Dianol, Dolmi)	PH4.127
p-Hydroxyphenylglycine (Athenon, Glycin, p-Hydroxyphenylaminoacetic Acid, Iconyl, Monazol)	PH4.128
p-Aminophenol Hydrochloride (Kodelon, P.A.P.)	PH4.129
Pyrogallic Acid (1, 2, 3-Trihydroxybenzene, Pyro, Pyrogallol	PH4.130
Catechol (Ortho-Dihydroxybenzene, Pyrocatechin, Pyrocatechol)	PH4.131

FSC 6750

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:13Z
Check the source to verify that this is the current version before use.

Column 1	Column 2 USA Standards
p-Phenylenediamine (1, 4-Benzenediamine, 1, 4-Diaminobenzene)	PH4.132
p-Phenylenediamine Dihydrochloride (1, 4-Benzenediamine Dihydrochloride, 1, 4-Diaminobenzene Dihydrochloride)	PH4.133
Chlorohydroquinone (Adurol, Chloroquinol, CHQ, Monochlorohydroquinone, 2-Chloro-1, 4-Dihydroxybenzene)	PH4.134
p-Benzyl-Aminophenol Hydrochloride (Duratol, Mono-Benzyl-para-Aminophenol Hydrochloride, N-Benzyl-p-Aminophencl Hydrochloride)	Ph4.135
1-Phenyl-3-pyrazolidone (Phenidone, Phenidone A, Graphidone)	PH4.136
Class III - Hardeners	
Aluminum Potassium Sulfate	PH4.150
Chromium Potassium Sulfate	PH4.151
Formaldehyde, 37-Percent Solution (Formalin)	PH4.152
Paraformaldehyde (Trioxymethylene)	PH4.153
Aluminum Chloride Solution	PH4.154
Class IV - Miscellaneous	
Sodium Sulfate, Anhydrous	PH4.175
Sodium Acetate, Anhydrous	PH4.176
Sodium Thiocyanate	PH4.177
Isopropylamine, 50-Percent Aqueous Solution (Monoisopropylamine)	PH4.178
Sodium Citrate, Dihydrate	PH4.179
Copper Sulfate (Cupric Sulfate)	PH4.180
Benzyl Alcohol	PH4.181
Ammonium Chloride	PH4.183
Ammonium Sulfate	PH4.184
Ethylenediaminetetraacetic Acid and its Salts (EDTA)	PH4.185
Hydroxylamine Sulfate (Hydroxyl-Ammonium Sulfate)	PH4.186
Hydroxylamine Hydrochloride	PH4.187
Ethylenediamine	PH4.188
Class V - Restrainers and Antifoggants	
A. Inorganic	
Potassium Bromide	PH4.200
Potassium Iodide	PH4.201
Potassium Chloride	PH4.202
Sodium Chloride	PH4.203
Sodium Bromide	PH4.207
B. Organic	
Benzotriazole (1, 2, 3-Benzotriazole)	PH4.204
5-Methylbenzotriazole	PH4.205
5-Nitrobenzimidazole Nitrate	PH4.206
Class VI - Alkalies	
Sodium Hydroxide	PH4.225
Potassium Hydroxide	PH4.226
Sodium Carbonate, Monohydrate	PH4.227
Sodium Carbonate, Anhydrous	PH4.228
Potassium Carbonate	PH4.229
Sodium Tetraborate, Decahydrate (Borax, 10 Mole)	PH4.230

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:13Z
Check the source to verify that this is the current version before use.

Column 1	Column 2 USA Standards
Sodium Metaborate, Octahydrate	PH4.231
Ammonium Hydroxide	PH4.232
Sodium Tetraborate, Pentahydrate (Borax, 5 Mole)	PH4.233
Sodium Phosphate, Tribasic	PH4.234
Class VII - Fixing Agents	
Sodium Thiosulfate, Anhydrous (Anhydrous Hypo)	PH4.250
Sodium Thiosulfate, Crystalline (Crystal Hypo)	PH4.251
Ammonium Thiosulfate, (60-Percent Solution) (Ammonium Hypo Solution)	PH4.252
Ammonium Thiosulfate (Ammonium Hypo)	PH4.253
Class VIII - Sulfites	
Sodium Sulfite, Anhydrous	PH4.275
Sodium Bisulfite, Anhydrous	PH4.276
Potassium Metabisulfite	PH4.277
Class IX - Bleaching Agents	
Potassium Dichromate (Potassium Bichromate)	PH4.300
Potassium Permanganate	PH4.301
Potassium Ferricyanide	PH4.302
Potassium Persulfate	PH4.303
Sodium Ferrocyanide	PH4.304

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards of the issues in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

Federal Specifications:
PPP-C-300 - Chemicals, Liquid, Packaging and Packing of.
PPP-P-16 - Packaging of Photographic Dry and Liquid Chemicals, General Specification for.

Federal Standard:
Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402.)

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.)

(Federal Government activities may obtain copies of Federal Specification, Standards, and Handbooks and the Index of Federal Specification and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specification and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:13Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Federal Caustic Poison Act.

(Information as to the availability of the Federal Caustic Poison Act may be obtained from the Department of Health, Education, and Welfare, Washington, D. C. 20201.)

United States of America Standards Institute Standards listed in 1.2.2.

(Application for copies should be addressed to the United States of America Standards Institute, 10 East 40th Street, New York, N. Y. 10016.)

3. REQUIREMENTS

3.1 Requirements of chemicals listed in column 1, paragraph 1.2.2, covered by this specification shall conform to the requirements of the United States of America Standards Institute Standards listed in column 2, paragraph 1.2.2.

3.2 Workmanship. Workmanship shall be in accordance with manufacturing techniques and practices covering manufacture of photographic chemicals as indicated in the applicable United States of America Standards Institute Standards.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Manufacturer. The manufacturer shall perform the sampling and inspection specified herein prior to submission of the product to the Government for acceptance.

4.2.1 Test equipment and inspection facilities. The manufacturer shall furnish and maintain all test equipment and inspection facilities for performing all inspection, and shall maintain complete inspection records. Test equipment and inspection facilities shall be of sufficient accuracy, quality, and quantity to permit performance of the required inspection. The manufacturer shall establish the accuracy and provide for the calibration of test equipment to the satisfaction of the Government. Manufacturers not having test equipment and inspection facilities satisfactory to the Government will engage the services of a commercial laboratory acceptable to the Government.

4.2.2 Inspection and test data. Records of control tests made to determine that each batch of a given photographic single substance chemical has been accurately formulated shall be kept by the contractor and made available to the Government.

4.3 Government. Acceptance of any chemical product covered by this specification shall be based upon verification by the Government of the manufacturer's compliance with the requirements, and sampling and inspection procedures of this specification. The Government may, at its option, repeat any or all of the inspections specified herein and in applicable referenced specifications or standards of the United States of America Standards Institute Standards.

4.4 Sampling. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.4.1 Batch. A batch will depend on the method of manufacture. For continuous modes of manufacture, a batch shall be defined as the entire quantity produced during one production cycle or one eight-hour period, whichever occurs first. For other than continuous modes, a batch shall consist of that amount of finished chemical resulting from one reaction mixture "charge."

4.4.2 Lot. For the purpose of sampling, a lot shall consist of all the chemicals of the same type, manufactured as one "batch" and offered for acceptance at one time.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:13Z
Check the source to verify that this is the current version before use.

4.4.3 Samples from lots. At the discretion of the procuring activity, samples of any or all lots of chemicals being supplied under contract shall be furnished without additional cost to the Government. These samples, consisting of one or more unopened containers, shall be selected at random from the material being supplied under contract, by the contractor, or by the Government inspector or authorized representative of the procuring activity, whoever is specified. These samples will be used in performing verification inspection. Samples consisting of unit packages, shall be selected at predetermined intervals, to assure sampling from the start, the finish, and the midway point, when appropriate, of each production run.

4.4.3.1 Sampling for inspection of filled container. A random sample of filled containers shall be selected from each lot in accordance with MIL-STD-105 at inspection level I, and at acceptable quality level (AQL) 2.5 percent defective, to verify compliance with all stipulations of this specification regarding fill, closure, marking, and other requirements not involving tests.

4.4.3.2 Sampling for tests. When practicable, for purposes of analysis, at least one original unopened container shall be taken at random from each lot segregated for inspection. When this is not practicable, a representative composite sample shall be taken, placed immediately in a clean dry glass bottle, sealed, marked, and submitted for testing as specified in 4.5.4. The sample shall be not less than two pounds nor more than 100 pounds.

4.5 Quality conformance inspection.

4.5.1 Inspection conditions and methods. Unless otherwise specified in the required test of the applicable referenced specification or standard of the United States of America Standards Institute, all inspections shall be made at ambient room temperature, pressure, and humidity.

4.5.2 Inspection of filled containers. Each sample-filled container selected in accordance with 4.4.3.1 shall be examined for defects of the container and closure, for evidence of leakage, and for unsatisfactory markings. Each sample filled container shall also be weighed to determine the amount of the contents. Any container in the sample having one or more defects, or under required fill, shall be rejected.

4.5.3 Inspection of preparation for delivery. An inspection shall be made to determine that the packaging and packing and marking comply with the requirements in section 5 of of this specification. Defects shall be scored in accordance with table I. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an acceptable quality level (AQL) of 4.0 defects per hundred units.

TABLE I. Classification of preparation for delivery defects

Examine	Defects
Marking (exterior and interior)	Omitted; incorrect; illegible, improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.
Contents (exterior and interior)	Number per container is more or less than required. Net weight exceeds requirements.

4.5.4 Inspection tests. Samples selected in accordance with 4.4.3.2 shall be subjected to the tests specified in the applicable standard of the United States of America Standards Institute (see 1.2.2, colum 2). If any sample fails in one or more of the applicable tests, the entire lot represented by the sample shall be rejected.

4.5.5 Rejected lots. Disposition of defective units or rejected lots shall be in accordance with MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 202640-26T11:13Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Photographic chemicals in quantities as specifed (see 6.2), shall be packaged as specified for level A in PPP-P-16 and PPP-C-300 as appropriate.

5.1.2 Level B. Photographic chemicals in quantities as specified (see 6.2), shall be packaged as specified for level A in PPP-P-16 and PPP-C-300 as appropriate.

5.1.3 Level C. Photographic chemicals shall be packaged in uniform quantities with sufficient outage provided in unit containers to prevent leakage or distortion due to expansion of contents through temperature changes occuring in transit from supplier to initial destination.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Photographic chemicals shall be packed in accordance with the requirements of level A of PPP-P-16 and PPP-C-300 as appropriate.

5.2.2 Level B. Photographic chemicals shall be packed in accordance with the requirements of level B of PPP-P-16 and PPP-C-300 as appropriate.

5.2.3 Level C. Photographic chemicals shall be packed to insure carrier acceptance and safe delivery at destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or order, the shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to markings required by contract or order, the interior and the shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 The chemicals covered by this specification are intended for use in photographic processing.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and should specify the following in procurement documents:

(a) Number, title, and date of this specification.
(b) Total weight required and size or weight of individual containers.
(c) Selection of the applicable level of packaging and packing required (see 5.1 and 5.2).
(d) Marking, shipping, and transportation description (see 5.3).
(e) Responsibility for inspection (see 4.1).
(f) Technical literature required.
(g) Additional marking in accordance with the Federal Caustic Poison Act, as required.
(h) Manufacturer's name and batch number.

6.3 Photographic grade. The term "photographic grade" refers to chemicals in which impurities known to be photographically harmful are limited to a safe quantity, while inert impurities are restricted to amounts not reducing the required assay strength.

MILITARY INTERESTS:

Custodians:
Army-EL
Navy-AS
Air Force-70

User activity:
Army-ME

Review activities:
Navy-AS
Air Force-70
Other-NSA, GS

Preparing activity:
GSA-FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 10 cents each.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:13Z
Check the source to verify that this is the currentversion before use.

GENERAL SERVICES ADMINISTRATION - FEDERAL SUPPLY SERVICE

SPECIFICATION COMMENT SHEET

BUDGET BUREAU NO. 29-R0175

INSTRUCTIONS

This form provides a way for users of this specification to inform the originator of problems encountered in its use. It is not to be used to request changes to accommodate proprietary features. All comments will be considered and appreciated, but please do not expect a reply. To comment: detach, complete, fold, staple, and mail.

NOTE: Comments on this form do not constitute or imply authorization to waive any part of the document or serve to amend contractual requirements.

1. SPECIFICATION

O-C-275C Chemicals, Photographic "Single Substance", General Specification for

2. CONTRACT NO. *(If any)*	3. QUANTITY ON CONTRACT *(Optional)*	4. DOLLAR VALUE *(Optional)*

5. GENERAL NATURE OF PROBLEM *(e.g., inspection difficulties, manufacturers unable to meet tolerances, containers collapse under normal warehousing conditions, etc.)*

6. SPECIFIC REQUIREMENTS AFFECTED *(Include paragraph number and lines of wording)*

7. SPECIFIC PROBLEMS *(e.g. tests in 4.2.2 will not assure that the battery will last required time; temperature ranges in table 2 do not conform to commercially available items.)*

8. RECOMMENDATIONS

9. NAME OF MANUFACTURER, ASSOCIATION, GOVT., AGENCY, ETC.	10. ADDRESS *(Number, Street, City, State and Zip Code)*

11. NAME AND TITLE OF SUBMITTER	12. DATE

GSA DC 69-9761

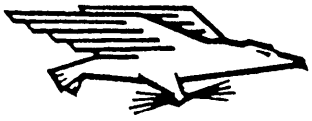
FORM GSA FEB 69 2200

Source: https://assist.dla.mil -- Downloaded: 2026l0-26T11:13Z
Check the source to verify that this is the currentversion before use.

FOLD

GENERAL SERVICES ADMINISTRATION

OFFICIAL BUSINESS



POSTAGE & FEES PAID
GENERAL SERVICES ADMINISTRATION

General Services Administration
Federal Supply Service FMSE
Washington, D. C. 20406

FOLD

GSA FORM FEB 69 2200

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:13Z
Check the source to verify that this is the current version before use.